Sahaj Bhadja

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Summary

Software Developer with 2+ years building **distributed systems**, concurrent data pipelines, multi-tenant databases, and production **GenAI applications** using Java, Python, Kubernetes, vector retrieval, cloud infrastructure, and rigorous testing.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Distributed Systems, Database Systems, Analysis of Algorithms, Operating Systems

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Data Structures, Database Management, Software Engineering

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, California*

- Architected a multi-tenant data platform using Azure SQL and Blob Storage with row-level isolation and normalized schemas for 500+ users, reducing analytical dashboard latency from 1.2 seconds to 350 milliseconds.
- Restructured fault-tolerant asynchronous GenAI orchestration using FastAPI background workers, processing 1K+ jobs with explicit state transitions and failure isolation while tripling throughput under concurrent workloads reliably in production.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, California*

- Engineered a multimodal GenAI platform using GPT-4o, Claude, Qwen-VL, and OCR, evaluating 10K+ daily submissions with 90%+ grading consistency while supporting switchable model workflows across production-scale educational workflows.
- Designed concurrent batch processing with multithreading and secure multi-format ingestion through REST APIs and AWS RDS, executing 1K+ files per batch with fault-tolerant recovery under peak load.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, California*

- Built a GPT-based Decision Transformer for offline reinforcement learning, developing end-to-end pipelines over 100K+ trajectory logs and improving long-horizon planning performance 18% through simulation-driven model evaluation at scale.
- Architected a spatial-temporal encoder integrating HD-map geometry and ego-agent history through multi-head attention, improving closed-loop planning stability 15% while reducing off-road behavior in simulation rollouts across challenging scenarios.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer, SecureSync *India*

- Optimized large-scale API integrations using object-oriented patterns, AWS Glue, and S3, improving response times 30%, increasing ETL pipeline efficiency 68%, and reducing technical debt 20% across production workloads.
- Engineered cross-platform data-sharing software with automated builds and 20+ unit and integration test suites, validating synchronization across three operating systems while collaborating with Microsoft and GitHub engineers.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern, VibrantChem *India*

- Implemented eight MERN microservices for authentication, permissions, billing, resources, customers, leads, and notifications, reliably synchronizing 2M+ records through REST APIs and event-driven WebSocket updates across distributed workflows.
- Automated Docker deployments through Jenkins CI/CD with unit, integration, and end-to-end validation, achieving 95%+ successful builds while integrating monitoring and secure release checks across production environments with reliable rollbacks.

Skills

Languages & Systems: Java, Python, C++, JavaScript, TypeScript, SQL, Bash, Linux, Object-Oriented Programming, Concurrency, Multithreading, Data Structures, Algorithms
GenAI & Retrieval: Retrieval-Augmented Generation, LLM Systems, Semantic Search, Vector Search, Embeddings, FAISS, Pinecone, Transformers, PyTorch, NLP, Multi-Head Attention
Data & Services: PostgreSQL, MySQL, SQL Server, MongoDB, NoSQL, Azure SQL, FastAPI, Flask, Node.js, REST APIs, Microservices, WebSockets, ETL Pipelines
Cloud & Engineering: Kubernetes, Docker, AWS S3, AWS RDS, AWS Glue, Azure, GCP, Jenkins, GitHub Actions, CI/CD, Git, JIRA, Postman, Unit Testing, Integration Testing, Debugging, Monitoring, Performance Optimization, System Design, Distributed Systems

Projects

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP*

- Built a retrieval-augmented document platform with secure ingestion, semantic vector search, real-time summarization, and collaborative workflows, maintaining 95% query accuracy across 50K+ documents for 200+ users with low-latency retrieval.

Chat2SQL Code | *Java, TypeScript, SQL, PostgreSQL, Docker, Maven*

- Built a Java and TypeScript database workspace supporting 10+ database systems, combining schema exploration, SQL generation, synchronization, and cross-platform deployment through Docker, Maven, Node.js, and Yarn for developer productivity.

CPU Scheduling Simulator Live | *C++, Algorithms, Operating Systems, React.js*

- Implemented six scheduling algorithms with interactive controls, dynamic waiting-time and turnaround-time calculations, and Gantt visualization, rigorously testing correctness across 1K+ preemptive and non-preemptive process combinations and workload distributions.